Efficiency of Machine Learning Methods for Extreme Precipitation Analysis

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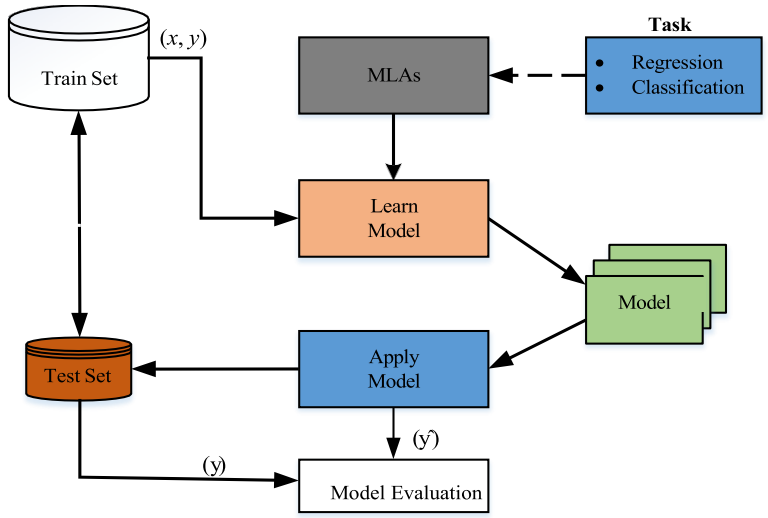
Introduction & Objectives: Over the past decades, many mathematical and computational methods such as numerical weather methods (Vitart and Robertson, 2018), and statistical and probability-based methods (Sapsis, 2021) have been used to analyze and forecast extreme climate events. Literature has shown that such traditional methods have various drawbacks including accuracy, weakness in uncertainty analysis, high computation cost, and the need for a comprehensive amount of data (Akpoti et al., 2019). Machine learning and deep learning methods have been able to tackle these limitations through their efficient computation and intelligence. Recently machine learning algorithms have been applied to solve or simulate multiple processes or key variables involved in the hydrological cycle, such as precipitation, air temperature, wind speed, radiation, evapotranspiration, soil moisture, and runoff (for example, Oppel and Fischer 2020; Zhao et al. 2019). Hence, there is a need for comparative performance analysis of several machine learning models for classification and regression tasks on climate extreme datasets This study aims to experimentally assess the performance of three machine learning models on extreme precipitation datasets for classification and regression tasks.

Methods & Data:



Three machine learning models used for this study are listed below:

- *Support Vector Machine*
- *Random Forest*
- *Artificial Neural Network*

Figure 1: Study Framework

Study Area and Data: This project focuses on the eastern United States that covers 2 U.S. National Oceanic and Atmospheric Administration climate regions – Northeast and Southeast. *i) Rain gauge precipitation data sets:* Station-based daily precipitation observations are derived from the NCDC U.S. Historical Climatology Network for the years 1951–2022. *ii) Gridded precipitation data sets:* The PRISM algorithm (available at http://www.prism. oregonstate.edu/)

combines point data with a digital elevation model to generate gridded estimates of precipitation along with a suite of climatological variables such as temperature, snowfall, and degree dew point among others. Data is available on the daily, monthly, and annual scale and at various spatial resolutions (800m to 4 km). In this work, the daily precipitation estimates are used (data set AN81d: PRISM Technical Note 2013).

Data preparation and Modeling: Figure 1 shows the models' training and testing framework for this study. Each clean dataset will be partitioned into two: 80% (train set) and 20% (test set).

Metrics for assessing the robustness of results.
The performance of the selected machine learning models will be compared using well-known and accepted evaluation metrics as shown in Table 1.

Table 1: Evaluation Metrics

Evaluation Metric	Equation	Description
R-squared (R^2)	$R^2 = 1 - \frac{RSS}{TSS}$	*RSS* = sum of squares of residuals *TSS* = total sum of squares
RMSE	$RMSE = \sqrt{\left(1/n \sum_{i=1}^{n} (y - \hat{y})^2\right)}$	y is the true value, $\hat{y}$ is the model's predicted value, n is the number of observations

Expected Outcomes:

- Many physical variables at different temporal and spatial scales such as temperature, relative humidity, soil moisture may be involved in determining extreme precipitation intensity and frequency. It is important to understand the variables most relevant to the extreme precipitation, which would help improve the predictability of extreme precipitations for near-term operation and long-term planning. ML models will be developed with inclusion of such physical variables as independent variables.
- Comparison of different machine learning models from this study will elucidate the efficiency and the potential use of machine learning approaches for the forecasting and classification of precipitation extremes.

References:

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